



Removal of Basic Blue 7 from Mesoporous Lignin Derived from Rice Husk

Lab Report

Submitted in partial fulfilment of the requirements for the degree of

Bachelor of Technology

(Biotechnology)

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DECLARATION

We hereby declare that the project entitled “Removal of Basic Blue 7 from Mesoporous Lignin Derived from Rice Husk” is an authentic record of our own work carried out at the School of Bioengineering and Biosciences, Lovely Professional University, Phagwara, for the partial fulfillment of the award of Bachelor of Technology in Biotechnology under the guidance of **Dr. Ajay Kumar.**

This work is our original work and has not been submitted for any degree/diploma in this or any other University. The information furnished in this report is genuine to the best of my knowledge and belief.

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CERTIFICATE

This is to certify that Sameer Roul (12113058), Shristi Tanaya (12113403), and Vullaganti Jayendra Kumar (12112915) are undergoing the project, entitled **“Removal of Basic Blue 7 from Mesoporous Lignin Derived from Rice Husk”** Under my guidance and supervision. To the best of my knowledge, the present work is the result of their original investigation and study.

No part of the report has ever been submitted for any other degree at any University. The reports are fit for submission and the partial fulfilment of the conditions for the award of B. Tech. Biotechnology.

Date: 28/04/2025

Supervisor Signature:

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Topic: Removal of Basic Blue 7 from Mesoporous Lignin Derived from Rice Husk

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1. INTRODUCTION

Synthetic dyes are widespread effluent pollutants, mostly caused by the textile, paper, food, and cosmetic industries, and they create severe environmental and health risks. They are manufactured in hundreds of thousands of tons every year, a large percentage of which are released into the waste stream. The colorful pollutants create a massive obstacle to light penetration even at ultra-trace concentrations in aquatic ecosystems, which inhibit photosynthesis and interfere with the delicate balance of living aquatic creatures (Mishra et al., 2023). Apart from their degradative products, most of these dyes are toxic, mutagenic, and carcinogenic, thereby posing direct threats to human and animal life through the food chain and waterborne pathogens. The major impact of these dyes is that they reduce the attractiveness of water, increase BOD and COD levels, and add more stress to the aquatic habitats. As a result, the demand for effective and sustainable solutions for removing dyes from industrial effluents has become a universal concern during the past few decades (Belpaire et al., 2015).

Biological treatment, chemical oxidation, membrane separation, coagulation/flocculation, electrochemical processes, and adsorption are some of the physical, chemical, and biological methods used to decolorize and detoxify dye-containing wastewater (Al-Tohamy et al., 2022). Biological processes are undoubtedly favored in an ecological context; however, they have limitations such as poor degradation rates and sensitivity to the toxicity of some dyes. While these chemical oxidation processes are sufficient for most dyes, they can also create secondary, toxic pollutants (Rane & Joshi, 2021). Separation by membrane technique has rendered excellent removal efficiency; however, in its implementation, fouling of membranes and operational costs are often hurdles to its use. Coagulation and flocculation could be used to remove suspended solids and, in part, certain dyes, but produce a large volume of sludge that must be further treated and disposed of (Khouni et al., 2020).

Lignin is a complicated heteropolymer aromatic compound, the second most abundant natural polymer, forming a substantial part of lignocellulosic biomass, which includes wood, agricultural waste, and energy crops (Nabipour & Hu, 2022). Lignin is available in vast amounts and constitutes a renewable, cost-effective resource today, as a byproduct of numerous industrial processes, including pulp and paper manufacturing and biorefinery operations (Poveda-Giraldo et al., 2021). The extensive chemical structure of lignin, with many functional groups like phenolic hydroxyl, carboxyl, and methoxyl groups, allows it to interact with dye

molecules via a variety of mechanisms, including electrostatic interactions, hydrogen bonding, and van der Waals forces. As a result, lignin has recently gained popularity as a bio-based adsorbent capable of adsorbing both heavy metals and organic dyes. Efforts have been made to convert and transform lignin into nanoparticles to improve its adsorptive characteristics (Kaur et al., 2023). With its reduced particle size and increased surface area, nano-lignin is a superior adsorbent compared to bulk lignin, from the viewpoint of excellent adsorption capacities, fast kinetic rates, and high accessibility of active binding sites (Das et al., 2023). Nano-lignin is produced using methods ranging from mechanical processes (milling and high-pressure homogenization) to chemical processes such as solvent extraction, enzymatic treatments, and controlled precipitations, among others (Perera et al., 2022).

Soda pulping, an alkaline pulping technique, has been extensively used in cellulose extraction from lignocellulosic biomass. Sodium hydroxide is used in the cooking process to solubilize lignin from cellulose fibers under alkaline conditions. Lignin is produced from the soda pulping process and is generally underutilized, thereby forming an important waste stream (Torres et al., 2020). The other use of soda lignin is for making a dye-removing nano-lignin. Model changes in the structure of lignin under the said soda pulp alkaline condition are changing the surface charge and reactivity of lignin, which in turn is causing a different interaction with dye molecules later during the adsorption process (S et al., 2023).

The study focuses on exploring the possibility of using lignin—an adsorbent obtained from post-soda pulp process waste—as a treatment system for textile dyes from aqueous solution according to sustainable waste management principles. This approach is designed to add value to a largely underutilized industrial waste and encourage greener alternatives for water purification (Bhatia et al., 2021). It is expected that the unique features of the structure and surface of this bio-nanocomposite may utilize the low-cost approach of hazardous dye removal for textile companies through soda pulping that dissolves lignin and converts it into nano-lignin (Xu & Ferdosian, 2017). It will comprise synthesis and characterization of the nano-lignin composite, systematically investigating of the major parameters that influence dye adsorption (namely pH, contact time, adsorbent dosage, and initial dye concentration, etc.), and finally various adsorption isotherms, kinetics, and thermodynamics to elucidate mechanisms of action of dye removal (Ismail et al., 2024). Fig 1 depicts Overview of the processes of lignin extraction and dye removal.

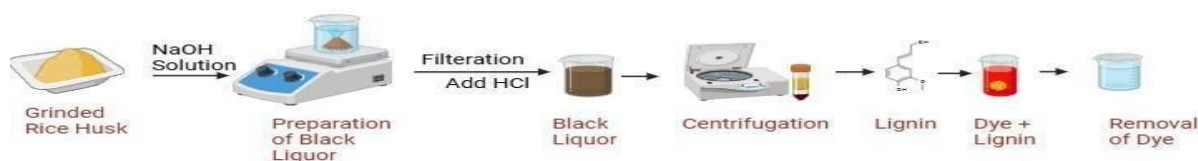


Fig 1: Overview of the processes of lignin extraction and dye removal

2. REVIEW OF LITERATURE

The presence of dyes in effluents generated by industrial sectors such as textiles, paper manufacturing, and food processing constitutes a major threat to the environment, as these chemicals can be toxic, carcinogenic, and capable of limiting light penetration and photosynthetic activity in aquatic ecosystems. Their removal from wastewaters has, therefore, been among the major challenges faced over the last few decades, and a great number of researchers have directed their attention toward the subject (Lellis et al., 2019). Color removal technologies include biological treatment, chemical oxidation, membrane separation, coagulation/flocculation, and ion exchange, among the most popular ones. Adsorption is said to be among the easiest, most efficient, and least expensive methods due to its high efficiency and capability to separate various pollutants (Razzak et al., 2022).

Several adsorbents, such as activated carbons derived from rice husk, sugarcane bagasse, wood sawdust, and agricultural wastes, have been studied to improve adsorbent efficiency based on key parameters including initial dye concentration, adsorbent dosage, contact time, pH of the solution, and temperature. These parameters are normally optimized in a batch mode (Dehghani et al., 2023). The majority of experimental data on adsorption fits the most widely used isotherm models, Langmuir and Freundlich. The above helps to measure the amount that can be adsorbed and characterize the surface properties of the adsorbent. On the other hand, the kinetics study is usually performed using pseudo-first-order and pseudo-second-order kinetics, implicating the rate of uptake and mechanisms of dye uptake (Chen et al., 2023).

With an expectation for the future development of lignin-based materials for dye adsorption, recent exploration of low-cost adsorbents for the removal of dyes has been generated from agricultural waste and biomass that have lignin as their principal component (Chahal et al., 2024). Apart from this, lignin is part of lignocellulosic material and has functional groups of hydroxyl and possibly carboxylic that could interact with dye molecules. Modified lignins such as carboxymethyl-lignin-alginate have been tested for their dye-removal capabilities from reactive blue MX-R; it has been indicated that it has an excellent adsorption capacity (Du et al., 2023). Similar to other adsorbents, a lignin-based adsorbent is influenced in its efficacy by the optimization of various critical factors, such as pH, contact time, and initial dye concentration. Knowledge of adsorption isotherms and kinetics is essential to understanding the various mechanisms of interaction between lignin-based materials and dye molecules. In addition, the present study aims to compare the dye-removal effectiveness of lignin-based adsorbents with commercial adsorbents such as activated charcoal to evaluate the feasibility of these adsorbents as one of the sustainable and economically feasible alternatives for dye wastewater treatment (Pourbaba et al., 2024).

3. RATIONALE AND SCOPE OF THE STUDY

The polluting activity of dyes in industrial effluents has grown tremendously in industries such as textiles, paper, and food. It has become a major environmental issue associated with toxic effects, carcinogenicity, and hindrance of light penetration and photosynthesis in aquatic ecosystems. There exist many processes to remove dyes; however, adsorption is considered a low-cost and versatile option. Lignin is a component of lignocellulosic materials. Agricultural waste and biomass to produce adsorbents commonly contains lignin, which makes sugarcane bagasse a low-cost adsorbent for dye removal. Thus, this gives credence to investigating lignin for dye removal. The objective of this research work is to address perennial pollution problems by investigating the effectiveness of either lignin-derived adsorbents or those modified with lignin for dye removal from aqueous solutions. Scope of the study- This would involve formulating and characterizing lignin-based adsorbents and evaluating their performance in batch adsorption for dye removal. Major operating parameters such as initial dye concentration, adsorbent dosage, contact time, and solution pH would determine this. It would also examine adsorption mechanisms using isotherm and kinetic models to analyze the experimental data. Subsequently, it would evaluate the practical application of lignin-based adsorbents regarding regeneration and reusability as well as the comparative performance of these adsorbents to the synthetic adsorbents.

4. OBJECTIVE OF THE STUDY

There is an ongoing concern for the environment resulting from the increased discharge of dye effluent, since most of these industrial contaminants are hazardous, often carcinogenic, and visible, and this directly decreases the penetration of light, affecting photosynthesis and also lives in aquatic environments. Considering that synthetic dyes are used in a wide range of industries, including textiles, paper, food, and others, there is a need to develop cost-effective and efficient methods of removing these dyes from wastewater. Adsorption stands out among the host of treatment technologies available today because of its simplicity and high efficiency in the removal of a wide range of dyes, coupled with the availability of low-cost and environmentally friendly materials. Thus, considerable research has focused on the application and introduction of environmentally friendly adsorbents from a range of options, including agricultural wastes and natural polymers, for the efficient removal of dyes. This study aims to establish the potential of lignin, a naturally occurring and inexpensive renewable biopolymer, as a method for developing an effective adsorbent for removing synthetic dyes from aqueous solutions. Lignin is present in the walls of all plant cells; it also plays a crucial role in the by-products of pulp and the paper industry and biorefineries. It has some intrinsic structural properties because of different functional groups, such as hydroxyl, carboxyl, and methoxyl groups. Due to these intramolecular, intermolecular, or hydrogen bonding, these groups can interact with dye molecules through electrostatic interactions or van Der Waals forces. The overall aim of this study is to determine the feasibility and efficiency of removing selected anionic and cationic dyes from aqueous solutions using lignin-based materials-ground in natural, chemically modified, or physically modified form. The main objective is accomplished through a series of experimental studies to understand the key factors that influence the dye removal process and determine whether lignin is a viable alternative to completely sustainable dye adsorption. Initially, the research studies would involve preparing and characterizing adsorbents based on lignin. Most likely, lignin would have to come from different sources, while other modification strategies, such as chemical activation or physical treatments aimed at the improvement of surface area, porosity, and concentration of active binding sites of the material, are explored. Meanwhile, lignin-based composites, quite possibly those with magnetic nanoparticles or other functional polymers, should also be synthesized and characterized for added functions like separation ease or enhanced selectivity toward specific classes of dyes. The characterization exercise of the lignin-based material would include quite extensive surface morphology analysis using a scanning electron microscope (SEM), surface

functional group determination by Fourier transform infrared spectroscopy (FTIR), and specific surface area and pore size determination by using the Brunauer-Emmett-Teller (BET) method. Next, the actual study will systematically screen the major process conditions affecting the adsorption performance of lignin-based adsorbents. These parameters include the initial solution pH affecting the surface charge of the adsorbent and the speciation of dye molecules, dosage of adsorbent determining the number of binding sites available, time allowing adsorbent and dye interaction or adsorption regulating kinetics of the adsorption process and attaining equilibrium, initial concentration of dye inducing mass transfer driving force and saturation capacity onto the adsorbent, and temperature affecting adsorption. Adsorption studies include different isotherm models such as Langmuir and Freundlich, and also other models to analyze equilibrium adsorption data to derive maximum adsorption capacity as well as to understand the adsorbent surface nature and its process. Furthermore, adsorption kinetics pseudo-first-order and pseudo-second-order models for evaluating the rate at which dyes are taken up, and could reveal the potential rate-limiting step processes, such as film diffusion and intraparticle diffusion during the adsorption-photolysis process. Lastly, the study will examine the possibility of regenerating and reusing these lignin-based adsorbents numerous times from the viewpoint of sustainability and cost-effectiveness. The emphasis will be placed on desorption methods based on various elution or thermal treatment methods to strip the absorbed dye molecules off the adsorbents and restore the original adsorption capacity for other possible adsorption-desorption cycles. These considerations provide a rounded view, giving way to lignin-based materials acting as effective, sustainable, and inexpensive adsorbents in treating anionic and cationic dye-polluted wastewater and thus facilitating the evolution of greener wastewater treatment technologies. Successful demonstration of lignin dye-removal capacity would open up huge areas of application for this abundant biopolymer, which goes hand-in-hand with being a greener alternative to synthetic adsorbents currently in use vis-à-vis environmental sustainability and resource utilization.

5. ABOUT BASIC BLUE 7

Basic Blue 7 is often referred to as Victoria Blue BO, C.I. Basic Blue 7, or C.I. 42595. It is a triarylmethane dye with the formula $C_{33}H_{40}ClN_3$, weighing approximately 514.14 g/mol. Basic Blue 7 appears as a reddish-blue or brownish powder or crystal. It easily dissolves in water and alcohol. It exhibits different colors in various chemical environments: a brownish-yellow color in concentrated sulfuric acid, red and yellow when diluted, and an olive-green color in concentrated nitric acid. It mainly dyes anionic substrates, such as wool, silk, nylon, and acrylics, in very bright violet-blue shades.

In contrast, cotton requires pretreatment with tannin for practical dyeing. In addition to its applications in textiles, Basic Blue 7 is used in microscopy staining (especially for mitochondria), printing inks, and as a smoke color for blue pyrotechnics; its relevance has also been studied in the context of cancer treatment, particularly in photodynamic therapy. Toxicologically, it exhibits antimicrobial properties but is known to cause contact allergies, resulting in restrictions on cosmetic applications in Europe, where safe concentrations have been established at 0.2% in hair dye products due to the risk of sensitization. Chemical syntheses are carried out through condensation reactions between bis(4-(diethylamino)phenyl)methanone and N-ethylnaphthalen-1-amine, along with phosphorus and amine chlorides. Basic Blue 7 is also known for its photosensitizing ability, with a density range of 1.05 to 1.2 g/mL and a log Kow value of 4.06, indicating moderate lipophilicity. Fig 2 and 3 depicts the 2D and 3D chemical structure of Basic blue 7, respectively.

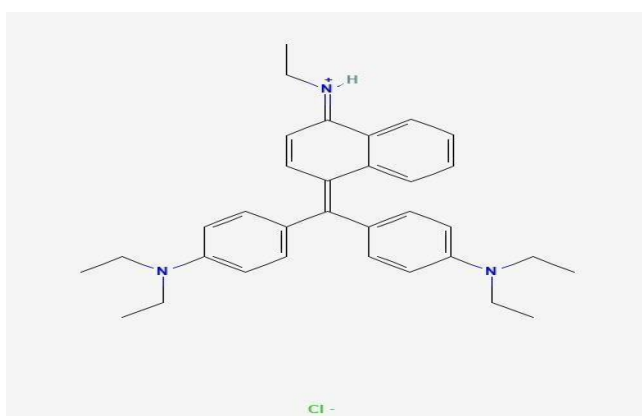


FIG 2:- 2-D Chemical structure ([PubChem CID 75444](#))

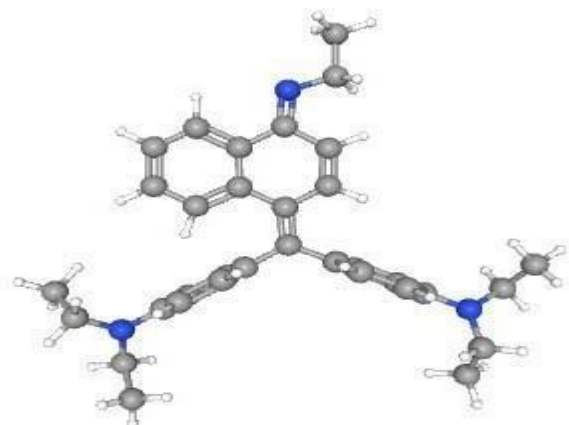


FIG 3:- 3-D Conformer (PubChem CID 75444)

5.1 IMPACTS OF DYE:

Health and ecological consequences follow the way of Basic Blue 7. It is toxic upon ingestion; upon exposure, it irritates the skin, severely damages the eyes, and irritates the respiratory tract (Lellis et al., 2019). Prolonged or repeated exposure might cause skin irritation, while ingestion may lead to gastrointestinal disturbances, including nausea, vomiting, and diarrhea. Environmentally, Basic Blue 7 is highly toxic to aquatic organisms, exhibiting long-lasting toxic effects that pose significant ecological risks, particularly in marine ecosystems (Visciano et al., 2016). It is said to be persistent in water, sediment, and soil, although it is not environmentally persistent in terms of bioaccumulation. The final regulatory evaluation indicates that Basic Blue 7 may cause both immediate and long-term impairments to the biological diversity of the environment (Wang et al., 2024). Toxicologically, it has been shown to induce photodynamic cytotoxicity in vitro and can kill human leukemic cells at very low concentrations when exposed to light. Its genotoxicity and reproductive/developmental toxicity are not established; however, it is a potential candidate for carcinogenicity based on the behavior of structurally related substances (Awasthi et al., 2024). Fig 4 explains the impacts of dye on the human body.

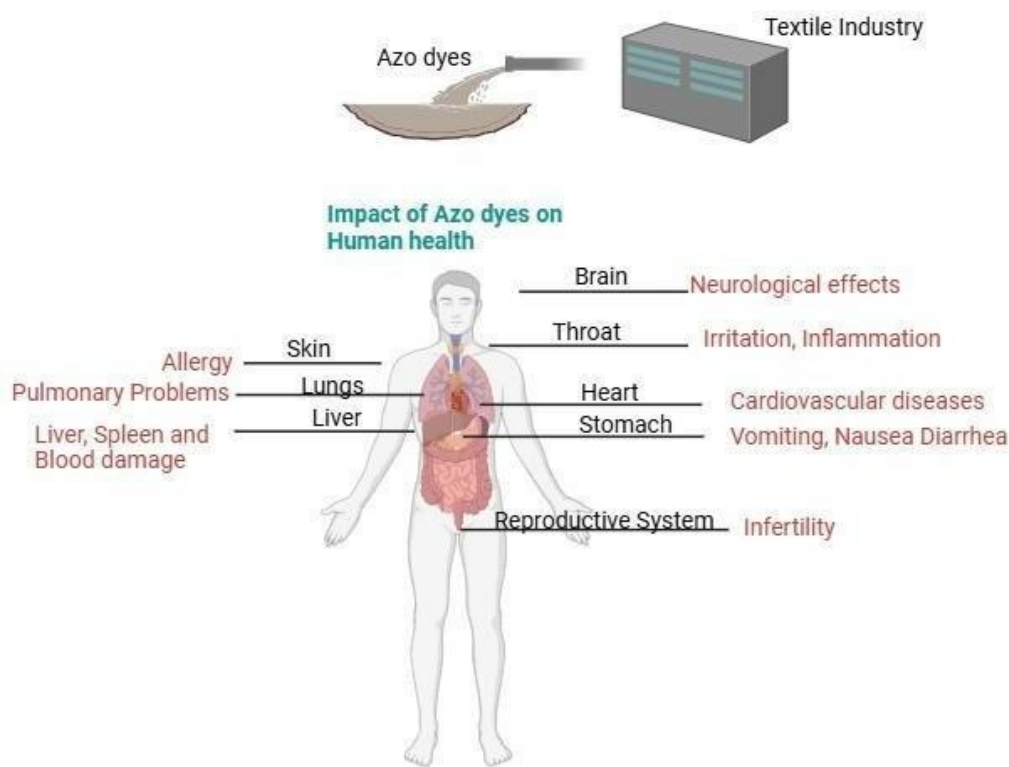


FIG 4:- Impacts of dye on Human Health

6. METHODOLOGY:

6.1 Soda pulping method

Soda pulping, one of the oldest and most economical chemical fractionations for lignocellulosic biomass, exploits the strong nucleophilicity of hydroxide ions to cleave lignin ether and ester bonds, releasing cellulose and hemicellulose fibers. Thus, the pretreated rice husk, high with lignin content ranging from about 20 to 30% (by dry weight), being at a very low cost as an agricultural by-product, served as the feedstock for cooking in aqueous NaOH solvent at concentrations of 5-15% w/v under elevated heat temperatures of about 150-200 °C for rapid penetration of caustic liquor into the porous biomass matrix and a very effective depolymerization of the lignin-echinocarpal complex (Dagnino et al., 2018). Thus, the combination of a moderate to high alkali charge and thermal severity solubilizes lignin fragments into the black liquor while swelling the remaining cellulose to improve access in downstream purification (Nagieb & Odeyemi, 1991). After most of the cooking, the brown stock is drained and washed to remove maximum residual lignin, and then screened to separate fibers. This is followed by feeding the spent liquor rich in dissolved lignin and hemicellulose

oligomers, and also entrained silica, to a recovery train with micro- and ultrafiltration modules to concentrate the lignins (Balasubramanian & Venkatachalam, 2023), acid precipitation towers for flocculation of phenolic fragments (Dagnino et al., 2018), and centrifuges followed by air drying to produce the high-purity, low-ash lignin powder for further applications. Meanwhile, concentrated black liquor is combusted in recovery boilers to regenerate sodium hydroxide (NaOH) (Nagieb & Odeyemi, 1991).

6.2 Raw materials required

6.2.1 Rice Husk

With the milling of rice, over 120 million tonnes of rice hull are produced each year, which is the protective outer covering of the rice grain. It is lignocellulosic biomass consisting of cellulose (35-40%), hemicellulose (20-25%), and lignin (20-30%) on a dry weight basis, with the actual lignin content typically reported as ranging from 20% to 30% of the total. One peculiar aspect of rice husk lies in its high silica content, which varies between 15% to 20%, that is deposited in its outer layers and contributes favourable and difficult properties (Nagieb & Odeyemi, 1991). The presence of silica provides a ready option for deriving high-value silica-based materials, such as nano-silica and silicates, from rice husk. In contrast, silica also presents a huge processing challenge, causing equipment abrasion during milling and greatly hindering enzymatic accessibility during biochemical conversions. The strong association of silica and the lignocellulosic matrix presents another complication for conventional pretreatment methods, whether acids, alkaline treatment, or biological methods, and might also require more stringent or tailor-made ones for the efficient fractionation and downstream valorization. However, notwithstanding these drawbacks, the uniqueness of silica in rice husk keeps it attractive for various applications such as biofuel, activated carbon, polymer composites, and adsorbents, provided that its silica-rich nature is dealt with during the processing.

6.2.2 Sodium Hydroxide

In contrast, the usual delignifying agent in soda pulping is sodium hydroxide (NaOH). Cooking is done at temperatures ranging from 140 °C to 170 °C, this is when NaOH infiltrates the cell wall matrix and cleaves the aryl ether (β -O-4) and ester linkages binding lignin to hemicellulose and cellulose into smaller soluble fragments that stay behind in the alkaline pulping liquor (black liquor) for due recovery. Hydroxide ions (OH^-) support nucleophilic

attack on the ether bonds of lignin. Simultaneously, such an alkaline environment promotes the ionization of the phenolic hydroxyl groups, further enhancing the solubilization and depolymerization of lignin (Dagnino et al, 2018).

The concentration of NaOH is an important parameter for soda pulping, as insufficient concentration causes incomplete delignification, resulting in dark, poorly separated fibers. Excess concentration causes too much degradation of carbohydrates, which lowers pulp yield and causes weakening of fiber strength. In practice, alkali charges for NaOH 5% (w/v) are adjusted according to the type of raw material (hardwoods vs. non-wood fibers, such as rice husk), the desired kappa number, and the pulp quality (Dagnino et al., 2018). For softwoods or agricultural residues with less lignin, lower alkali concentrations (5-8%) would suffice, but for more recalcitrant materials, either a higher load (10-15%) or additives (anthraquinone, hydrogen peroxide) are required in order to achieve satisfactory delignification without excessive loss of carbohydrates. After the cooking stage, the spent liquor containing lignin dissolved into the water, degraded hemicellulose, and residual NaOH is separated from the pulp, the solid phase is treated with wash water that removes black liquor, and the liquor is then concentrated and burned in a recovery boiler for the regeneration of NaOH (Dagnino et al., 2018).

6.3 Process Steps

6.3.1 Pretreatment of Rice Husk

Husk of rice is pretreated to eliminate impurities from its surface in order to achieve uniform chemical penetration into the biomass prior to the soda pulping process. Initially, the husk was placed in a washing beaker and washed with clean or distilled water, thus removing adhering soil, fine dust, or inorganic soluble salts. The washing facilitates a reduction in ash and silica content (Dagnino et al., 2018). The cleaned husk is then air-dried at a moderate temperature (60-90 °C). This drying reduces the moisture content, which otherwise would have weighed above 10-15%, to below 5%, thus further saving energy in the heating steps ahead, apart from promoting more uniform alkali uptake (Dagnino et al., 2018). Facilities subject rice husk to liquid hot water pretreatment. When rice husks were slurried in water at the ratio of 1:5 (i.e., 1 part of solid to 5 parts of water), the slurries were heated at temperatures ranging from 160 to 200 °C for a duration of about 10 to 30 minutes and allowed to cool back to room temperature. The mechanism of this heating may result in partial hemicellulose solubilization as well as in the loosening of lignin from the carbohydrate complex. This, in turn, increases pore volume

and exposes additional reactive sites to caustic liquor during pulping (Nagieb & Odeyemi, 1991). The next step involves the cooling of the husk, its dewatering, and re-drying down to the desired moisture content (below 5%); after this stage, the husk goes into soda pulping. Providing such simple washing and drying, as well as optional hydrothermal steps, would not only improve delignification efficiency through pretreatment but also enhance overall pulp yield and quality.

6.3.2 Soda Pulping

The pretreated rice husk, at the soda pulping stage, is charged using an aqueous sodium hydroxide solution in a 1000 mL beaker for chemical delignification to release cellulose and hemicellulose fibers from the lignin matrix. With so much hydroxide concentration at such extreme conditions of operation—150-200 °C, atmospheric pressure to some 10 Bar, and NaOH concentrations between 15 % (w/v)—hydroxide ions attack β -O-4 ether bonds and ester linkages of lignin, further breaking the polymer into smaller water-soluble fragments (Nagieb & Odeyemi, 1991). The reaction time itself has a wide range, from less than half an hour states for extremely high severity and high alkali conditions for continuous processes to several hours under milder conditions in batch operations for pulping. This, however, gives room for process engineers to balance out the extent of delignification (normally monitored using the kappa number) against carbohydrate degradation and yielding strength of the pulp at large (Nagieb & Odeyemi, 1991). Within the beaker, a homogeneous temperature and chemical environment are necessary. Agitation, under the assistance of a magnetic bead, mixes the rice husk particles with the solution uniformly. The constant heat supply is offered to the glass beaker over the magnetic bead stirrer (Balasubramanian & Venkatachalam, 2023). At the end of the cooking period, the heat is turned off, and the "black liquor" (pulp plus spent liquor) is discharged. The pulp rejects dissolved lignin and residual alkali from the washing process. The residual from this process is kept for further regeneration of chemicals. This integrated process not only has the potential to maximize the recovery of chemicals and improve energy efficiency but also minimizes environmental impact due to the closure of the chemical loop, as well as a decreased effluent load (Balasubramanian & Venkatachalam, 2023). Fig 5, Fig 6 depict black liquor and filtration of black liquor, respectively.

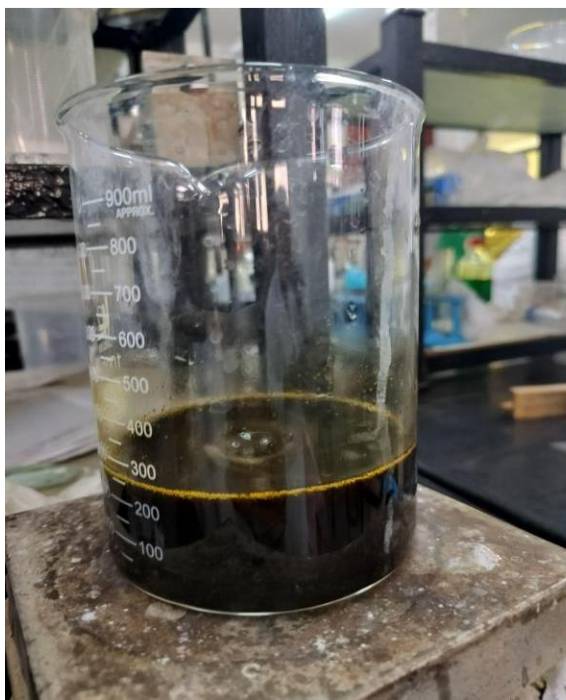


Fig 5: Black liquor



Fig 6: Filtration of Black liquor

6.3.3 Lignin Separation and Recovery

The thick black liquor, which is dark brown due to the presence of dissolved lignin fragments, hemicellulose oligomers, and colloidal silica, is first filtered using a woven filtration cloth with a pore size of around 50 μm to separate coarsely husked fibers and hefty cellulose fines (Nagieb & Odeyemi, 1991). Then, the clarified lignin-laden filtrate is subjected to acid precipitation, as this is the key step in lignin recovery. Stirred in a beaker with the aid of a magnetic bead, hydrochloric acid was used to lower the pH from the native alkaline value of approximately 12-13 to the optimal pH range of 2.0-3.0. Phenolic hydroxyl and carboxyl groups are protonated at $\text{pH} < 4$, resulting in the reduction of solubility in water and the formation of colloidal flocs. However, over-acidification reduces the pH to less than two, risking the co-precipitation of hemicellulose fragments along with an increase in inorganic salt content in the lignin cake (Nagieb & Odeyemi, 1991). Acid induction is performed slowly over 20-40 minutes under moderate agitation (100-300 rpm) and within a temperature range of 25-40 $^{\circ}\text{C}$, ensuring that flocculation kinetics are rapid. This approach balances higher temperatures against the potential for thermal condensation to alter the chemical functionality of lignin. (Dagnino et al., 2018).

Very tight control of pH, temperature, mixing intensity, and residence time enables the maximization of lignin yield, which accounts for 85-90% of the theoretical lignin content. In this manner, functional groups, such as aliphatic hydroxyls, methoxyl, and β -O-4 linkages, which determine the reactivity of recovered lignin in further applications, are preserved. After precipitation, the lignin flocs are separated by operators using high-speed disc or decanter centrifuges, which operate under centrifugal accelerations of 6000 to 10000 G. The wet cake generally contains 40-60% solids, depending on the centrifuge design and the solids concentration in the feed. The moist cake is subjected to extensive washing and drying to meet the required purity and moisture specifications for commercial lignin powder. In the first wash, the cake is slurried again with acidified water ($\text{pH} \approx 3\text{-}4$) in the solid-to-liquid ratio of 1:3-1:5 (w/v) and agitated at 100-200 rpm for 10-20 minutes to remove occluded hemicellulose oligomers, residual Na^+ , Cl^- or SO_4^{2-} ions, and entrained silica particles, bringing the ash content down below 5% (Dagnino et al., 2018). This wash liquor is decanted or centrifuged off, and the process is repeated two more times, with each wash using successively cleaner water, ensuring that the levels of fermentable sugars and inorganic salts are further reduced to less than trace levels ($< 1\%$ w/w). Deionized or soft water may be used in the final washing to further improve efficiency. Then, this washed cake is pressed in a filter press to obtain a 30-40% solids cake prior to drying. Then, this dewatered product goes through the drying process, which will convert the wet cake to stable powder with a moisture content usually lower than 5%. This low water content is essential for maintaining shelf life, improving flowability, and facilitating further downstream handling. By means of this method, the drying process involves the use of a hot air oven at $60\text{ }^\circ\text{C}$ for 2 hours. Finally, the product is then cooled, and agglomerates are sieved out. The product is then packaged either in inert atmosphere or desiccant-packed bags to avoid moisture uptake and oxidative changes, giving a high-purity and low-ash product that is readily applicable (Nagieb & Odeyemi, 1991). Fig 7 and Fig 8 depict lignin pellet after centrifugation and dried and powdered form of lignin, respectively.



Fig 7: Lignin pellet

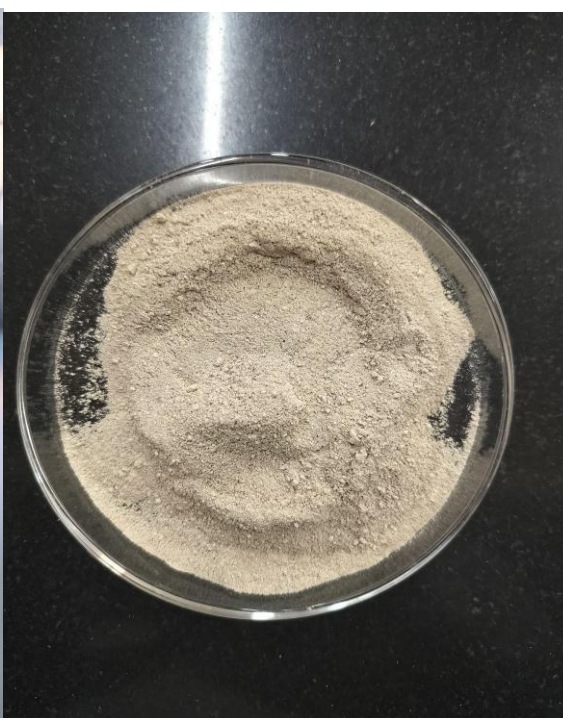


Fig 8: Powdered Lignin pellet

7. RESULT AND DISCUSSION

In the study mentioned in this article, Fourier Transform Infrared spectroscopy (FTIR) is thoroughly added as an analytical approach for the identity and confirmation of lignin in samples. FTIR is recognized in the scientific community for its ability to present complex molecular facts based on how molecules absorb infrared radiation at certain vibrational frequencies. IR radiation causes a sample to absorb characteristic frequencies via the practical agencies of the molecules, leading to characteristic peaks in the spectrum. According to biopolymer examples, these peaks act as molecular identifiers, permitting researchers to trace and differentiate among one-of-a-kind forms of biopolymers primarily based on their particular vibrational fingerprints. The FTIR spectra of the pattern indicated many excessive-depth peaks that were interpreted at the same time in favor of the lignin presence. A sharp and nicely described height inside the 1600-1590 cm^{-1} variety becomes one of such widespread observations. The band of absorption corresponds to the stretching vibrations of carbon-carbon double bonds ($\text{C}=\text{C}$) in aromatic jewelry, the structural building blocks of the polyphenolic lignin complex structure. The presence of this aromatic character is one highly differentiating element among lignin and different plant biopolymers, which include cellulose and hemicellulose, which typically lack fragrant jewelry and consequently do not have a corresponding vibrational band in this spectral vicinity.

The occurrence of this peak has been previously documented in the literature, with studies by Faix (1991) and Sharma et al. (2016) confirming its value as a characteristic marker or fingerprint tool for identifying lignin. Earlier studies have ensured that C=C aromatic stretching vibrations yield characteristic and reproducible spectral features, which serve in favor of FTIR as a sensitive technique for identifying lignin in complicated biological matrices. Besides the large peak, other observable absorption bands between 1510 and 1500 cm^{-1} , corresponding to aromatic skeletal bond vibrations, can be seen in the spectra. The occurrence of this peak has been documented within the literature, with studies by Faix (1991) and Sharma et al. (2016) confirming its value as a characteristic marker or fingerprint tool for identifying lignin. Earlier studies have ensured that C=C aromatic stretching vibrations yield characteristic and reproducible spectral features, which serve in favour of FTIR as a sensitive approach for identifying lignin in complicated biological matrices. Besides the peaks, other observable absorption bands between 1510 and 1500 cm^{-1} , similar to aromatic skeletal bond vibrations, can be seen in the spectra. From the evaluation of the sample, it is observed that the FTIR spectrum has specific and crucial functions that agree with the presence of lignin inside the substance considered (rice husk). Among these observations, special emphasis is laid upon the absorption bands exhibited between 1260 and 1230 cm^{-1} . These peaks characterize the vibrational and stretching modes of C–O that refer to the aryl ether linkages, that is an integral structural function of lignin's complex 3-dimensional network.

These ether linkages are very crucial as they structurally guide lignin's structure, linking aromatic units through ether bonds and supplying it with distinct three-dimensionality. This spectral signature, in particular the peaks related to aryl ether linkages, is one precise feature that permits lignin to be outstanding from different plant cellular wall components like cellulose and hemicellulose. These polysaccharides denote a paucity of such ether-type linkages as are located in lignin; cellulose has specific bands associated with its beta-1,4-glucan chain and scattered unique styles of adsorptions via hemicelluloses. Thus, figuring out C–O stretching vibrations in the location of 1260–1230 cm^{-1} is strong evidence of the presence of lignin. For verifying the validity of such an identification, the spectral data underwent rigorous validation. Numerical evaluation of the wavenumber positions and absorbance intensities had been achieved in order to set up reference spectra of pure lignin samples.

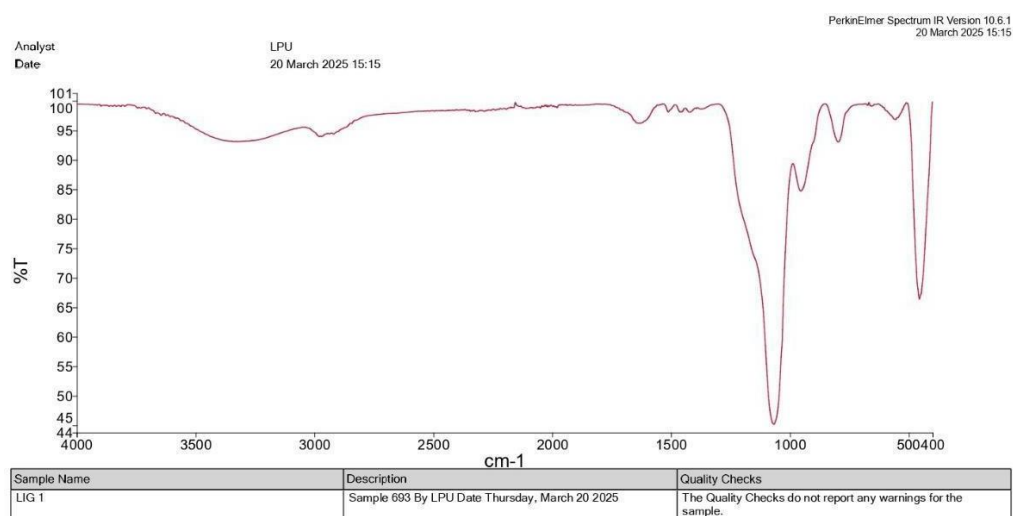
A close match to the peaks positions within the pattern spectrum with those of the reference pattern, with little deviation and close fit with absorbance intensities, suggests that there are identical concentrations and structures, and for this reason, reinforces the inference of lignin

within the analyzed sample. Furthermore, the proprietary software program supplied spectral representation that confirmed feature absorption bands around 1600 cm^{-1} due to aromatic C=C stretching vibrations and, in the place of $1510\text{--}1500\text{ cm}^{-1}$ associated with aromatic skeletal vibrations. Such peaks are the hallmark features of lignin's aromatic ring systems. Uniformity of these spectral features across multiple representations is proposed through the graphical plots and tabulated numerical values, backing up the reproducibility of the analysis. To summarize, integrating spectral proof, especially the peaks linked to aryl ether C–O stretching and aromatic ring vibrations, has resulted in the deduction of the presence of lignin in the examined sample. That the spectrum has been reputable attests to the soundness of this identification and gives a sturdy basis for further structural or compositional research.

Such an intensive method substantiates the feasibility of FTIR spectroscopy within the characterization of complex biopolymers, thereby allowing a stringent differentiation amongst plant cell wall elements in keeping with their signature vibrational transmission. As depicted by Faix (1991), FTIR features of lignin corroborate our findings in confirming the presence of peaks characteristic of aromatic rings and useful functional groups. This validates our findings, indicating that the spectral properties are consistent with established structures of lignin and furthers our analytic credibility. (Zhou et al., 2011; Sharma et al., 2016) report similar FTIR features for lignin isolated from poplar and other biomass sources, respectively. These studies thus attest to the peaks discerned in the FTIR spectra being consistent with those assigned to lignin, thereby reinforcing the case for its presence in the analyzed sample. FTIR spectral analysis has firmly established the presence of lignin in the sample.

Furthermore, among these absorption bands, prominent ones were observed near regions associated with aromatic C=C stretching ($\sim 1600\text{ cm}^{-1}$), aromatic skeletal vibrations ($1510\text{--}1500\text{ cm}^{-1}$), and C–O stretching in aryl ether linkages ($1260\text{--}1230\text{ cm}^{-1}$). Each one of these spectral characters is well described in the literature as typical of lignin, thus adding weight to the interpretation. A key point is that there was also no sign of strong absorptions in the regions typically associated with hemicelluloses or cellulosic polymers, such as around 1730 cm^{-1} (ester carbonyls) or around 1100 cm^{-1} (C–O–C stretches of polysaccharides). This suggests a minor contribution from non-lignin biopolymers and supports the lignin signal's high specificity. The spectral characteristics were consistent across the various form outputs and well correlated with literature-reported spectral profiles for lignin; thus, the reliability of the conclusions is enhanced. This indicates that the FTIR technique effectively detects lignin and

identifies several of its critical structural signatures. Identifying these lignin-specific bands provides a solid foundation for future analytical work, including quantifying lignin content and structural characterization. Additionally, revealing lignin at this stage is necessary for assessing applications related to biomass valorization and material development. Fig 9 is a graph of FTIR analysis of lignin



Page 1

Fig 9: FTIR of LIG 1 sample

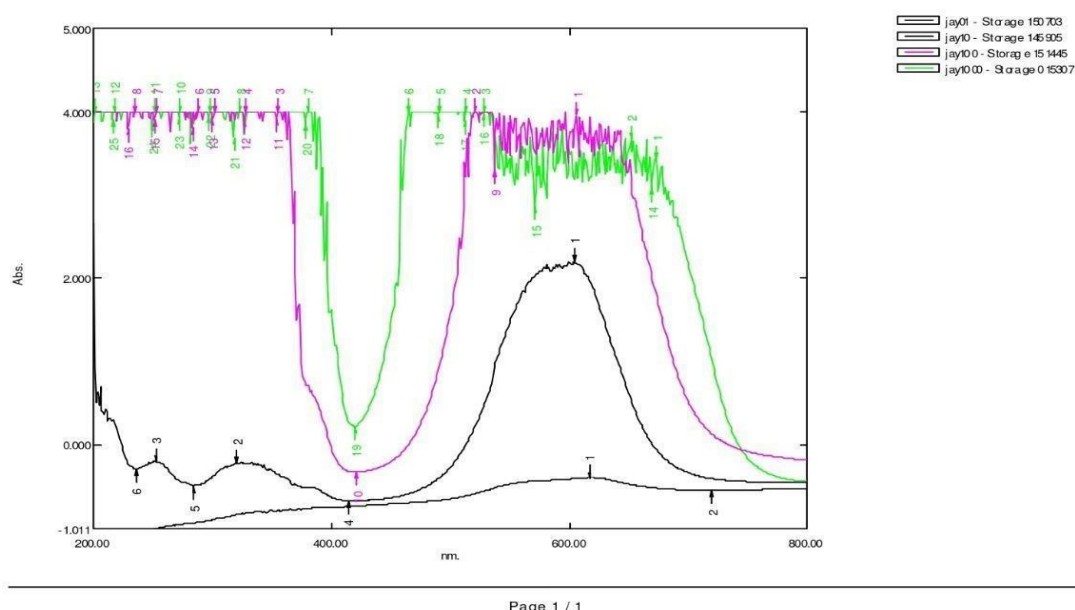


Fig 10: UV-Vis spectra of jay01, jay10, and jay100, (Basic blue 7).

Fig 10 depicts the data of UV vis spectrometry of basic blue 7 .This work employed ultraviolet-visible (UV-Vis) spectroscopy to assess the optical properties and storage-induced changes of the jay01, jay10, and jay100 samples, providing insight into the structural stability and electronic behavior of these samples. Overlaying spectra recorded from 200 to 800 nm exhibited dramatic differences between the samples, where jay01 (black curve) presented a wide and high peak absorption centering at about 600 nm, indicating that it is a stable conjugated system or bears plasmonic resonance characteristic of well-ordered structures. In comparison, jay10 and jay100 (pink and green curves, respectively) exhibited a sharp drop in absorbance at approximately 400 nm, followed by higher noise levels, which are inferred to be indicative of significant degradation and heterogeneity. The unperturbed smooth profile of jay01 suggested a very low level of aggregation or scattering. In contrast, the noisy baselines in jay10 and jay100 with fluctuating signals suggested high polydispersity, structural fragmentation, and possible oxidation. A hypochromic shift, accompanied by a decrease in absorbance intensity, was observed in samples stored under conditions that likely accelerated degradation processes, leading to the disruption of extended conjugation and loss of functional optical properties. These findings are supported by a comparative analysis with relevant literature in which comparable degradation behaviors have been reported in conjugated

polymers, lignin derivatives, and nanoparticulate systems, thereby underscoring the role of environmental stress in defining the integrity of a material. The dip around 400 nm in degraded samples may also indicate the formation of oxidation products or the breakdown of primary chromophores into smaller, non-conjugated fragments, which alter the electronic transitions responsible for light absorption. The present study examines the importance of robust storage protocols in protecting sensitive materials from oxidative environments and photodegradation, thereby preserving their structural integrity and functionality. Its findings are also far-reaching, particularly concerning the use of such materials in any field that requires long-term optical performance, such as sensors, optoelectronics, and catalysis, as this would lead to significant performance losses. The results from the UV-Vis analysis presented strong arguments for the material's sensitivity to environmental conditions. They emphasized the need for ongoing research in material stabilization techniques to enhance their robustness for practical applications.

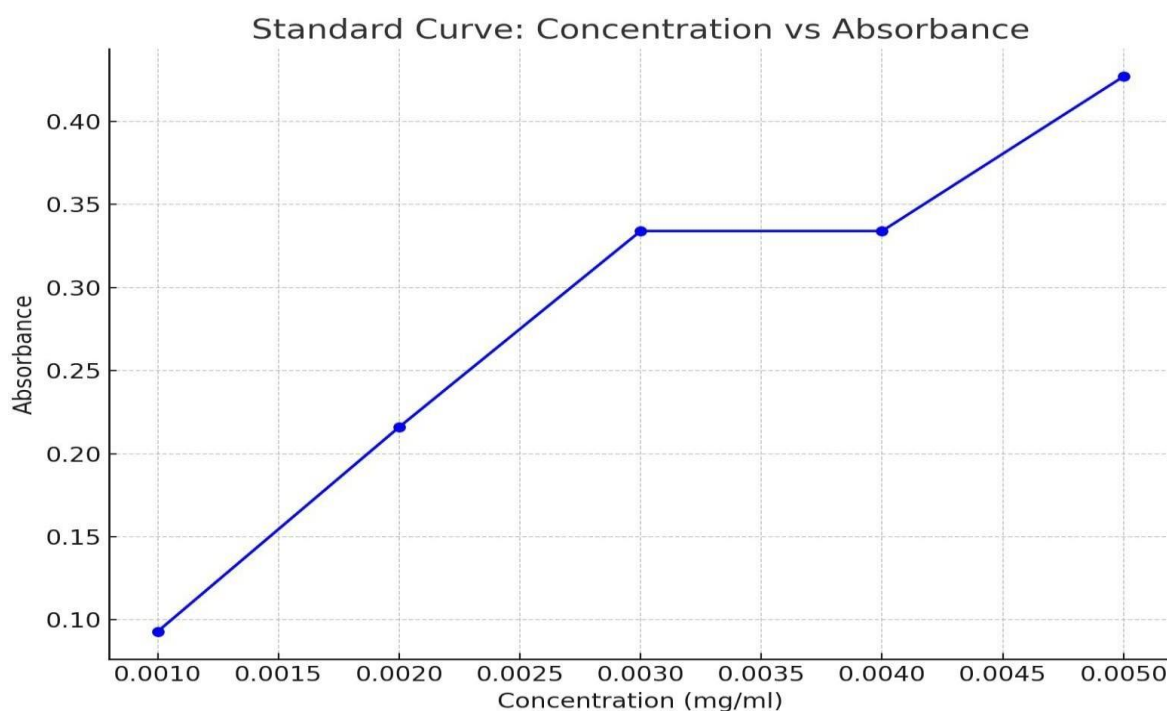


Fig 11: Standard curve of Basic Blue 7 at the visible spectrum

At the concentration of the dye, we measure the absorbance value at a specific wavelength (600 nm) and correlate it to create a standard curve. The standard curve provides a correlation between dye concentration and absorbance, from which unknown dye concentrations in experimental samples can be calculated with some accuracy from their corresponding absorbance readings. This is very much needed because absorbance directly relates to the

amount of dye present in solution, given a range within which Beer-Lambert's law holds. The standard curve also serves in adsorption studies by relating the amount of dye remaining in solution after the treatment of the adsorbent with, for example, lignin via experimental absorbance values compared to the calibration line. This, in turn, indirectly measures the amount of dye adsorbed onto the lignin surface. Without the standard curve, it would have been impossible to quantify adsorption efficiency or relate the effectiveness of the adsorption process to a specific set of conditions, such as pH. Hence, the standard curve is a fundamental tool for translating raw absorbance data into concentration values, enabling the accurate scrutiny of dye removal capability in scientific studies. Fig 11 depicts the standard curve of basic blue 7 at visible range.

This graph illustrates the impact of pH on the adsorption of dye molecules by lignin, in comparison to a standard calibration curve relating concentration to absorbance. The standard curve, indicated by the blue line, serves as a reference for calculating the amounts of dye based on absorbance readings. It exhibits close to linear behavior, with absorbance changing against increasing dye concentrations, and is suitable for measuring adsorption phenomena at different pH levels. Red markers were used to denote experimental absorbance values at three pH levels —pH 3, pH 4, and pH 5 —overlaid upon the standard curve. Each label corresponds to a specific data point, allowing for the comparison of adsorption efficiencies at different pH settings. Table 1 depicts the experimental values obtained for absorbance at different pH levels.

Experimental values obtained:

Concentration	pH	Absorbance
10 ppm	3	0.876
10 ppm	4	0.784
10 ppm	5	0.376

Table 1: Absorbance of dye by lignin at different pH levels

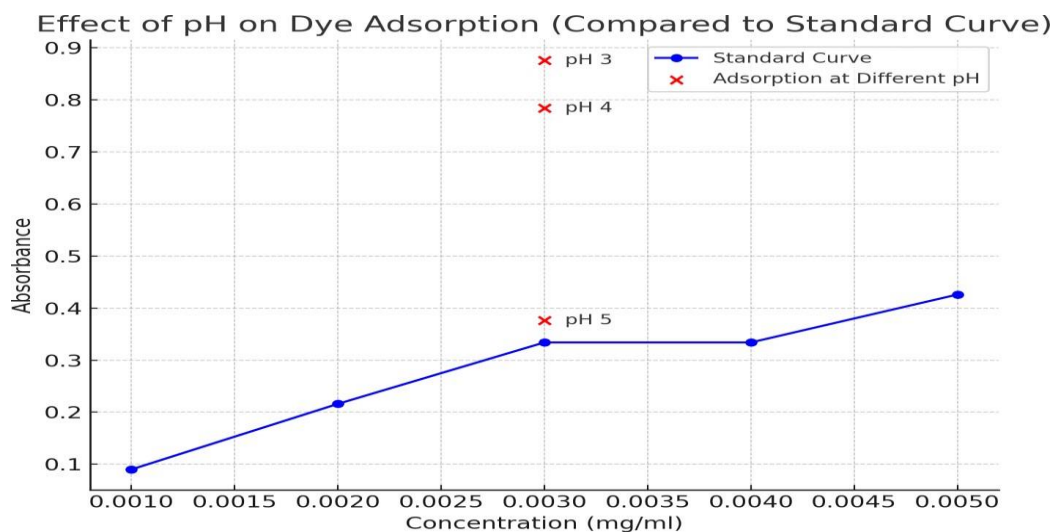


Fig 12: Effect of pH on dye adsorption

Fig 12 depicts the effect of pH on dye adsorption. The influence of pH on the dye-level adsorption efficacy on lignin is scientifically illustrated in the graph, which compares experimental absorbance values at different pH levels with the standard absorbance versus concentration curve. The blue straight line represents the standard curve, confirming the direct relationship between the dye concentration and absorbance. The superimposed red points indicate the absorbance values after adsorption at pH 3, 4, and 5. Low absorbance indicates very efficient dye removal from the solution, while higher adsorptions correspond to increasing absorbance. Among the various tested conditions, pH 5 exhibits the lowest absorbance, indicating maximum dye adsorption.

In contrast, absorbance gradually increases at pH 4 and pH 3, indicating lower dye removal and, consequently, lower adsorption performance in acidic conditions. The acidic attitude, therefore, suggests that the adsorption capacity for lignin strongly depends on the change in pH, likely due to changes in interaction scenarios resulting from variations in surface charge and the ionization state of the functional groups involved in the binding mechanism. Hence, the graph indicates that the dye adsorption efficiency is enhanced in mildly acidic environments, while excessive acidity impairs the dye removal function of lignin. Therefore, pH optimization is crucial for any dye binding process.

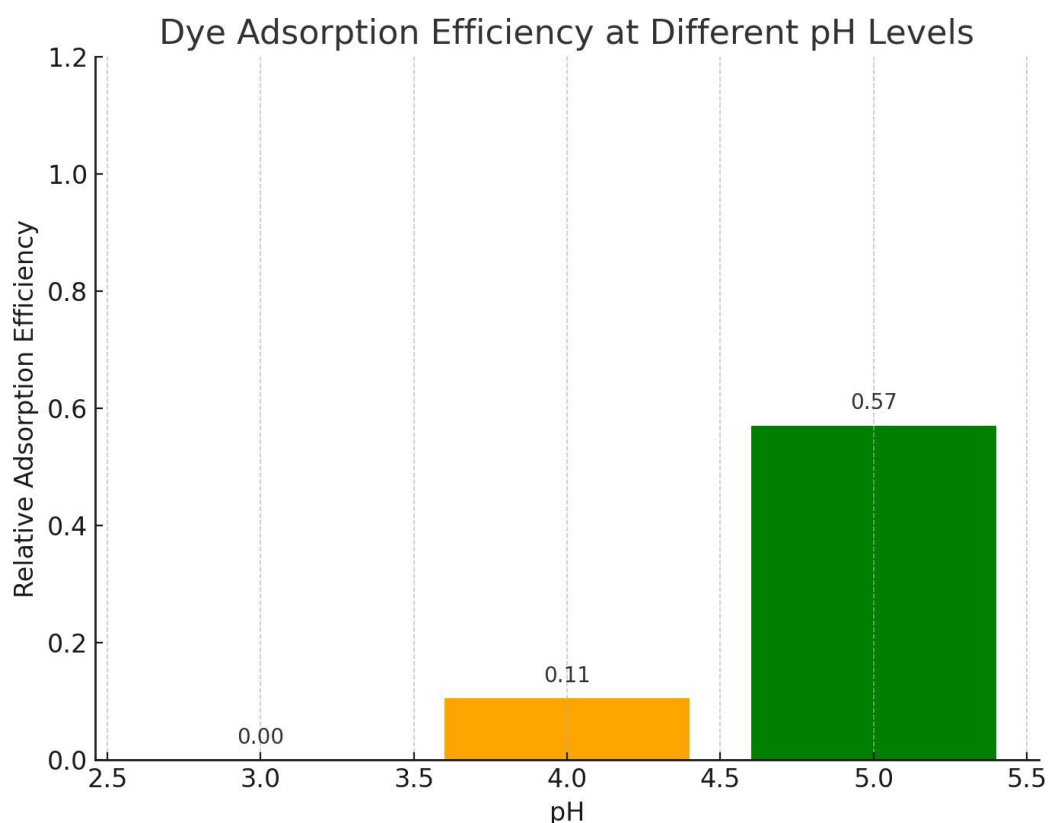


Fig 13: Dye adsorption efficiency at different pH

Fig 13 depicts the efficiency of dye adsorption at different pH. This graph represents the relative efficiencies of dye adsorption at various pH levels and describes the effect of pH on the effectiveness of dye removal by lignin. It indicates that the adsorption efficiency increases considerably as the pH varies from 3 to 5. At pH 3, the adsorptive efficiency is almost zero, showing little removal of dye in cases with a very high acidic condition. A small advance was made at pH 4, which is still less than efficient and indicates little adsorption activity. However, the most significant increase occurred at pH 5, where adsorption versus pH proved to be much more favorable under moderately acidic conditions. The bar colors also emphasize this observation, as it improves from minimal gray to low orange and then high green. Thus, dyeing under highly acidic conditions inhibits the functional groups of lignin that are responsible for binding the dye. With a slight increase in pH, they recover their activity, leading to improved adsorption. Hence, it is clear that the optimal aspects of the pH of the solution significantly impact the dye adsorption potential of lignin-based systems, with a pH of 5 among the conditions studied being the most suitable.

In this study, initial dye concentrations were kept constant for all the conditions under study. Upon the completion of the adsorption process, the dye concentration remaining in the solution was determined by measuring absorbance. Lower absorbance correlated with greater adsorption efficiency and a greater removal of dye from the solution phase, while high absorbance values indicated less dye uptake by the lignin substrate. The examination of the graph reveals that the absorbance at pH 5 is significantly lower than that observed at pH 4 and pH 3. This lends credence to the idea that dye adsorption, in terms of efficiency, occurs in a small acidic environment (pH 5). The subsequent increase in absorbance with a decrease in pH indicates that the adsorption capacity is diminished as the conditions become more acidic. Hence, conversely, the highly acidic conditions would prove unfavorable for dye adsorption, with efficiency declining significantly from pH 5 to pH 3. The cause of the behavior mentioned above can be explained based on the surface chemistry of lignin and the ionization behavior of the dye molecules. In the range of firm acidity, the excess hydrogen ions are likely responsible for the protonation of the functional groups in lignin, thereby reducing the available adsorptive sites. A lower pH may also alter the electrostatics of the dye, weakening any favorable interactions with the lignin surface.

In contrast, at pH 5, the lignin surface is more accessible for interaction with dye molecules, creating favorable conditions for enhanced adsorption performance. In unison, the findings emphasize the significant importance of pH in modulating dye-lignin interactions. Optimal adsorption occurs at a pH of 5 and decreases drastically under more acidic conditions. The practical implications for wastewater treatment include controlling pH exactly for enhanced removal efficiency of dyes. All of this demonstrates that the effectiveness of the adsorption process is improved, while also being more sustainable and economically viable for the entire lignin-based remediation.



Fig 14: Dye solution before adding lignin

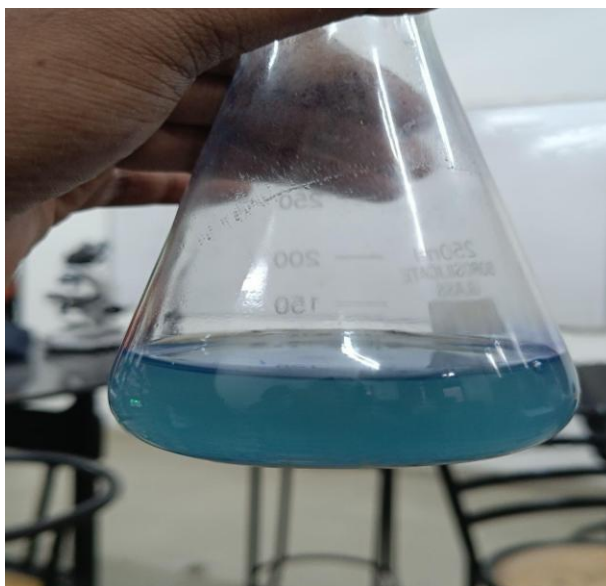


Fig 15: Dye solution after adding lignin

Figs 14 and 15 depict the dye solution before and after adding the lignin, respectively. The investigation into the adsorption of Basic Blue 7 dye on the lignin at pH 5 was conducted to evaluate the removal efficiency and subsequently to elucidate the mechanisms by which this interaction occurs between the dye molecules and the lignin surface. With the introduction of lignin, a pronounced change occurred in the appearance of the dye solution, where the intense deep blue coloration of the untreated dye solution was significantly toned down, instead rendering a bluish-green hue as lignin was added to the solution. This addition provided semi-quantitative evidence of the lignin's ability to capture Basic Blue 7 from the solution. Through qualitative observation and subsequent establishment and quantification of the adsorption behavior, we subjected our experimental data to the classical equilibrium models, namely the

Langmuir and Freundlich isotherms, to extract the characteristic parameters of adsorption capacity and affinity (Langmuir et al., 1918; Freundlich et al., 1906).

We unraveled the interaction of lignin with Basic Blue 7 dye at pH 5 in an attempt to elucidate details of its removal and the mechanistic bases for the possible dependent/dye interaction-controlling processes. The drastic change in the appearance of the dye solution, without the introduction of lignin, occurred when the intense, deep-blue dye solution shifted to look lighter and took on a bluish-green hue. The semi-quantitative proof that lignin was able to capture Basic Blue 7 from the solution was evident from the immediate color change. Beyond qualitative observation and moving to quantify the adsorption behavior, we applied two classical equilibrium models, known as the Langmuir and Freundlich isotherms, to fit our experimental data, from which we could extract parameters characterizing adsorption capacity and affinity (Langmuir et al., 1918; Freundlich et al., 1906).

The Langmuir model is based on the premise of monolayer adsorption onto a surface with a finite number of identical, energetically equivalent binding sites. Under this assumption, when a dye molecule occupies a site, no more molecules will be allowed to attach at that site, thus assuming that all sites are similar. The excellent fit of the model to the equilibrium data suggested that Basic Blue 7 molecules indeed tend to form a single layer on the surface of lignin (Langmuir et al., 1918). From this analysis, we derived two important values: the maximum amount of dye that lignin can adsorb under our experimental conditions, also known as saturation capacity, and a constant that measures the strength of attraction between the dye molecules and the lignin surface. Lignin was determined to have a saturation capacity of around 30 mg of dye per gram of lignin, indicating that lignin excels as a dye adsorbent. The affinity constant provides evidence of a relatively strong interaction between the dye and lignin, indicating that once dye molecules reach available sites, they are more likely to remain bound than to desorb back into solution. We applied the Freundlich model (Freundlich et al., 1906) not only to complement the monolayer assumption but also to counter it, treating adsorption site energies differently; some sites are strong, while others are weaker, thereby modifying multilayer adsorption possibilities. In this case, there might be a variable adsorption capacity, and the intensity of adsorption may change with concentration. From the data in this analysis, two empirical parameters are derived: one relating to the overall capacity of the lignin surface, and the second classifying or identifying the degree of favorable adsorption based on the ability to adsorb over a range of concentrations. Both parameters from our measurements indicated favorable processes; the overall capacity parameter was strong, while the intensity parameter

indicated that, as dye concentration increased, the effectiveness of adsorption also increased. Nevertheless, although the Freundlich model generally described the observations obtained in our experiments, it certainly did not correlate with the experimental data as closely as the Langmuir presentation did, particularly under the case of higher concentrations of dye, at which point the predictions of the Freundlich model would over-predict dye removal (Freundlich et al., 1906). This suggests that, despite some heterogeneity, lignin still behaves more uniformly as a surface under our set of conditions, where dye molecules occupy similar types of sites until a complete layer forms. These two isotherm analyses are complementary in establishing a consistent rationale for lignin interaction with Basic Blue 7 at pH 5. Under acidic conditions, many of lignin's functional groups, such as carboxylic and phenolic moieties, become protonated, thereby lowering the negative surface charge and weakening its electrostatic attraction towards the positively charged dye. Nevertheless, adsorbing is robust, which suggests that non-electrostatic interactions prevail. Chief among these is that the aromatic structure of lignin can stack itself onto the aromatic rings of Basic Blue 7, thereby mediating strong and stable bindings. Hydrogen bonding, as well as hydrophobic interactions, contributed to the retention of the dye, acting as a counterbalance to the lowered level of electrostatic contributions. Thus, the greater agreement of our data with the Langmuir model suggests that the dye molecule is attracted to distinct, energetically favorable sites on the lignin surface, packing into a single layer until no free sites remain. The calculated dimensionless factor, which indicates whether a given adsorption will occur under the given conditions, also provides further evidence that adsorption is favorable. For our case, this factor's value fell in the favorable range for dye uptake, thus reinforcing the conclusion that lignin is an effective adsorbent across all concentrations tested. This finding is highly relevant for practical applications, such as textile wastewater treatment, where dye concentrations can vary. It indicates that lignin will perform reliably and will not be easily saturated or inhibited by high dye loads. Therefore, high uptake capacity, favorable interaction parameters, and performance in acidic conditions offer prospects for lignin to become an economical and sustainable alternative to currently used adsorbents. This is in contrast to synthetic adsorbents such as activated carbon. Lignin, a byproduct of the paper and biofuel industries, is available in bulk amounts and is therefore very inexpensive. Since it is a renewable resource, it can also be subjected to chemical modifications or physically activated to maximize its performance for specific dyes or effluent streams. The performance of lignin at low pH creates an opportunity for scenarios in which wastewater effluents are naturally acidic and thus do not require any extensive pH adjustment before treatment. The clear fit to monolayer adsorption behavior from

the Langmuir analysis indicates that one can confidently extrapolate equilibrium data from other cationic dyes. There are matters to discuss regarding the heterogeneity and multilayer nature of the adsorption process, which indicate the necessity of surface modification of lignin to enlarge the binding energy spectrum. This might further assist in increasing capacity or honing selectivity against specific dye structures. The underlying technicalities of such modeling can aid in understanding batch reactor design, where dosing and mixing times are optimized through equilibrium capacity, or in continuous-flow conditions, such as in packed beds, where the prediction of breakthrough curves can be achieved through the same variables. These engineers could estimate the amount of lignin to be used to achieve a certain removal efficiency, how often the adsorbent would need to be replaced or regenerated, and what would be the required contact times with knowledge of saturation limits and affinity constants. Fig 14 and 15 depicts the dye solution before and after adding lignin into it.

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